



CODECHEF

The CodeChef.com Challenge, your monthly dose of coding puzzles from India's biggest coding contest, is now in print!

*H*ello and welcome to the *CodeChef Puzzle Challenge* sponsored by *LINUX For You* and *Direct!* Solve this month's puzzle, and get a chance to be one of the three lucky people who'll win a cash prize of Rs 1,000 each!

The February edition's puzzle

There are 10 baskets containing walnut cookies. In those, 9 baskets contain cookies, each weighing 10 grams, while one basket contains cookies, each weighing 20 grams. All the cookies look identical and there are plenty of them in all the baskets. Since we don't want to eat broken cookies, the baskets cannot be lifted or moved for weighing. There is a digital weighing machine which is allowed to be used only once. How will you find out which of the baskets contains the heavier cookies?

The solution

Number each basket from 1 to 10.

Select 1 cookie from Basket 1, Select 2 cookies from Basket 2, Select 3 cookies from Basket 3, and so on, i.e., 'n' cookie(s) from basket 'n', where $1 \leq n \leq 10$.

You will now have $(1+2+\dots+10) = 55$ cookies. If they all weighed 10 grams, then the weighing machine will show $55 \times 10 = 550$ grams.

But if the weight was, say, 570 grams, then you will come to know the number of cookies that weigh an extra 10 grams, $570 - 550 = 20$;

$20/10 = 2$

So, the basket with heavier cookies (in the above considered case) is Basket 2.

Generally, if the total weight turns out to be 'm' grams, then the basket number will be $(m-550)$ divided by 10.

And the winners this month are:

- Sathwik G N
- Logeshwaran
- Divya Choudhari

Your CodeChef Puzzle of the Month

Ram and Shyam sell donuts in the market. Ram sells jumbo donuts, at the rate of 2 for Rs 50 while Shyam sells normal-sized donuts at the rate of 3 for Rs 50. At a certain point when both have the same amount of donuts left, Shyam is called away. He asks Ram to take care of his donuts. To avoid making everything too complicated, Ram puts all the donuts in one basket, and starts selling at the rate of 5 donuts for Rs100. When Shyam returns at the end of the day, all the donuts have been sold. But when they start dividing the money, there appears to be a shortage of Rs 350. Suppose they divide the amount of money equally, how much does Ram lose with this deal?

Looking for a little extra cash?

Solve this month's *CodeChef Challenge* puzzle and send in your answer to codechef@fyindia.com or lly@codechef.com by March 15, 2012. Three lucky winners will get exciting prizes, both in cash and merchandise! You can also participate in the contest by visiting the online space for the article at <https://www.facebook.com/LinuxForYou> 

About CodeChef: CodeChef.com is India's first, non-commercial, online programming competition, featuring monthly contests in more than 35 different programming languages. Log on to CodeChef.com for the March Challenge that takes place from the 1st to the 11th, to win cash prizes up to Rs 20,000. Keep visiting CodeChef to participate in the multiple programming contests that take place on the website throughout the month!